Follow the moving relation and keep the original target

$F(2) = (2, 4) \bmod 8$
Moving row: $(-2, 1)$
$(-2, 1) \cdot (2, 4) = 0$

Leading layer mod 2
$F(2)/2 = (1, 2) \mapsto (1, 0)$
Residual Jacobian image

Original: $x^2 + 1 = 0 \bmod 4$
Squares are 0 or 1
No root

Augmented equation
$x^2 + 1 + y = 0 \bmod 4$
$(1, 2)$ works, but $y \neq 0$